

ROLANDO S. GARCIA, PH.D.

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EDUCATION

University of California, Berkeley Ph.D. in Computer Science advised by Dr. Joseph M. Hellerstein Dissertation: <i>The Management of Context in the Machine Learning Lifecycle</i>	Berkeley, CA 2017 – 2024
Arizona State University B.S. in Computer Science, <i>Summa Cum Laude</i>	Tempe, AZ 2014 – 2017
Northern Arizona University Philosophy and Computer Information Systems	Flagstaff, AZ 2010 – 2013

AWARDS AND FELLOWSHIPS

ASU Presidential Postdoctoral Fellowship, two years of funding.	2024
Dell Technologies Inc. Research Award, one year of funding.	2023
NSF Graduate Research Fellowship, three years of funding.	2017
UC Berkeley Chancellor's Fellowship, two years of funding.	2017
UC Berkeley's EECS Excellence Award.	2017
CRA Outstanding Undergraduate Researcher Award, honorable mention.	2017
Fulton Undergraduate Research Initiative, research award.	2016
IEEE VAST Grand Challenge Award, outstanding comprehensive submission.	2015

RESEARCH

School of Computing & AI Presidential Postdoctoral Fellow, advised by <u>Dr. Ross Maciejewski</u> • <i>Submitted and under review:</i> * NSF TTP-T: <i>Scalable Human-in-the-Loop Review: Expert-AI Collaboration in High-Volume Document Processing</i>, w/ R. Maciejewski, & T. Winarski¹ · Co-PI at 40% effort, \$759k over 3 years. * NSF Future CoRE: <i>III: Explainable GNN Approaches to Ad-Hoc Queries on Heterogeneous Networks</i>, w/ Yixuan He² et al. · Co-PI at 20% effort, \$1M over 3 years.	July 2024 – Present <i>Arizona State University</i>
RISE Lab rise.cs.berkeley.edu NSF Graduate Research Fellow, advised by <u>Dr. Joe Hellerstein</u> • FlorDB: a hindsight logging database for the machine learning lifecycle. • ... How Engineers Operationalize Machine Learning, w/ Shreya Shankar et al. • Internships: Linea Labs (<i>startup</i>).	June 2017 – June 2024 <i>University of California, Berkeley</i> github.com/ucbrise/flor
Amazon A9 (Search) Research Intern, Search Infrastructure Team • Maintained model training pipelines via AWS SageMaker, Docker. • Bulk data processing and regular re-training.	May 2018 – December 2018 <i>Palo Alto, CA</i>
VADER Lab vader.lab.asu.edu Undergraduate Researcher, advised by <u>Dr. Ross Maciejewski</u> • NSF REU: Visual Analytics Algorithms for Spatiotemporal Analysis. • Visual Analytics of Crowd Spatiotemporal Data (IEEE VAST Grand Challenge Award).	January 2015 – May 2017 <i>Arizona State University</i>

¹ Sandra Day O'Connor College of Law

² School of Mathematical and Natural Sciences

PUBLICATIONS

- Digitization as View Maintenance: From Source-Preserving Capture to Verified Corpora.** Erick Duarte, Akshit Jain, Rohan Boddu, Tyson Winarski, Ross Maciejewski, Rolando Garcia. *Conference on Innovative Data Systems Research*, under review. *Submitted, CIDR 2027*
- Citation Analysis and Deterministic Evaluation at Scale.** Rolando Garcia, Akshit Jain, Shiyi Liu, Rostyslav Hnatyshyn, Fan Lei, Ross Maciejewski. *ACM Transactions on Interactive Intelligent Systems*, under review. *Submitted, TIIS 2026*
- IDSPACE: A Novel Document Generator for Reliable Evaluation of Digital Identity Verification Systems.** Lulu Xie, Yancheng Wang, Kanchan Chowdhury, Rolando Garcia, Yingzhen Yang, Jia Zou. *IEEE International Conference on Data Mining*, under review. *Submitted, ICDM 2026*
- Cost-Benefit Modeling of Interactive Visual Machine Learning Systems.** Yixuan Wang, Rolando Garcia, Alex Endert, Ross Maciejewski. *IEEE Transactions on Visualization and Computer Graphics*, 2026. *VIS 2026*
- Flow with FlorDB: Incremental Context Maintenance for the ML Lifecycle.** Rolando Garcia, Pragya Kallanagoudar, Chithra Anand, Sarah E. Chasins, Erin M. Kerrison, Aditya G. Parameswaran, Joseph M. Hellerstein. *Conference on Innovative Data Systems Research*, 2025. *CIDR 2025*
- “We Have No Idea How Models will Behave in Production until Production”: How Engineers Operationalize Machine Learning.** Shreya Shankar*, Rolando Garcia*, Joseph M. Hellerstein, Aditya G. Parameswaran. *ACM Conf’ on Computer Supported Cooperative Work and Social Computing* 2024. *CSCW 2024*
- *Co-first authors. **Honorable Mention: MLOps Community Best Paper Award**
- Quilt: Custom UIs for Linking Unstructured Documents to Structured Dataasets.** Pragya Kallanagoudar, Chithra Anand, Rolando Garcia, Rebecca M. M. Hicke, Aditya G. Parameswaran, Eunice Jun, Sarah E. Chasins. *UIST Adjunct ’24: Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, 2024. *UIST Demo 2024*
- Hindsight Logging for Model Training.** Rolando Garcia, Eric Liu, Vikram Sreekanti, Bobby Yan, Anusha Dandamudi, Joseph E. Gonzalez, Joseph M. Hellerstein, Koushik Sen. *Proceedings of the VLDB Endowment*, 14(4): 682-693, 2021. *VLDB 2021*
- Context: The Missing Piece in the Machine Learning Lifecycle.** Rolando Garcia, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw, Joseph E. Gonzalez, Joseph M. Hellerstein. *Workshop on Common Model Infrastructure* at KDD, 2018. *KDD CMI 2018*
- A Visual Analytics Framework for Exploring Theme Park Dynamics.** Michael Steptoe, Robert Krüger, Rolando Garcia, Xing Liang, Ross Maciejewski. *ACM Transactions on Interactive Intelligent Systems*, 2018. *ACM TIIS 2018*
- The State-of-the-Art in Predictive Visual Analytics.** Yafeng Lu, Rolando Garcia, Brett Hansen, Michael Gleicher, Ross Maciejewski. *Computer Graphics Forum*, 2017. *CGF 2017*

TALKS

- Incremental Context Maintenance for the ML Lifecycle.** *Rolando Garcia, Pragya Kallanagoudar, Chithra Anand, Sarah E. Chasins, Erin M. Kerrison, Aditya G. Parameswaran, Joseph M. Hellerstein. Conference on Innovative Data Systems Research. January 2025.* *CIDR 2025 Amsterdam*
- How Engineers Operationalize Machine Learning.** *Shreya Shankar, Rolando Garcia, Joseph M. Hellerstein, Aditya G. Parameswaran. ACM Conf' on Computer Supported Cooperative Work and Social Computing. November 2024.* *CSCW 2024 Costa Rica*
- Flow with FlorDB: Agile MLOps without Compromises.** *Rolando Garcia. Colorado School of Mines. October 2024.* *Mines Seminar, 2024 Golden, CO*
- Operationalizing Machine Learning: An Interview Study.** *Rolando Garcia, Shreya Shankar. December 2022.* *Flyte, 2022*
- Operationalizing Machine Learning: An Interview Study.** *Shreya Shankar, Rolando Garcia. Berkeley, CA, October 2022.* *Inaugural Address, 2022 EPIC, UC Berkeley*
- Operationalizing Machine Learning: An Interview Study.** *Shreya Shankar, Rolando Garcia. October 2022.* *Canva, 2022*
- Operationalizing Machine Learning: An Interview Study.** *Shreya Shankar, Rolando Garcia. San Francisco Bay Area, October 2022.* *MLOps World, 2022 San Francisco*
- FLOR: Managing the Infinite Virtual Metadata of Model Training.** *Rolando Garcia. Cornell University, February 2022.* *Cornell Seminar, 2022*
- Hindsight Logging for Model Training.** *Rolando Garcia, Eric Liu, Vikram Sreekanti, Anusha Dandamudi, Bobby Yan, Joseph Gonzalez, Koushik Sen, Joseph Hellerstein. International Conference on Very Large Databases. August 2021.* *VLDB 2021 Copenhagen*
- Choose What to Log, After you Execute.** *Rolando Garcia, Eric Liu, Vikram Sreekanti, Anusha Dandamudi, Bobby Yan, Joseph Gonzalez, Koushik Sen, Joseph Hellerstein. Retreat talk to lab sponsors. Monterey, CA. January 2019.* *RISE Retreat 2019*
- ML in Context: Preserve, Share, and Analyze Experiments with Flor.** *Rolando Garcia, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw, Sona Jeswani, Eric Liu, Malhar Patel, Joseph Gonzalez, Joseph Hellerstein. Talk and Tutorial for lab sponsors. Berkeley, CA. October 2018.* *RISE Camp 2018*
- Context: The Missing Piece in the Machine Learning Lifecycle.** *Rolando Garcia, Vikram Sreekanti, Neeraja Yadwadkar, Daniel Crankshaw, Joseph Gonzalez, Joseph Hellerstein. The 24th ACM SIGKDD Intl' Conf. August 2018.* *KDD 2018 London*
- Data Management Challenges in the Machine Learning Lifecycle.** *Rolando Garcia, Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Sona Jeswani, Eric Liu, Malhar Patel, Joseph Gonzalez, Joseph Hellerstein. Invited talk at Amazon Search. Palo Alto, CA. April 2018.* *Amazon 2018 Palo Alto*

Design, Run, and Reproduce Experiments with Jarvis. <i>Rolando Garcia, Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Alexey Tumanov, Joseph Gonzalez, Joseph Hellerstein.</i> Retreat talk to lab sponsors. Monterey, CA. January 2018.	<i>RISE Retreat</i> 2018
Jarvis: Managing the Machine Learning Lifecycle. <i>Rolando Garcia, Vikram Sreekanti, Daniel Crankshaw, Neeraja Yadwadkar, Alexey Tumanov, Joseph Gonzalez, Joseph Hellerstein.</i> Invited talk at Google. Mountain View, CA. September 2017.	<i>Google</i> 2017 Mountain View
The State-of-the-Art in Predictive Visual Analytics. Yafeng Lu, <i>Rolando Garcia</i> , Brett Hansen, Michael Gleicher, Ross Maciejewski. 19 th <i>EG/VGTC Conference on Visualization</i> , Barcelona, Spain. June 2017.	<i>EuroVis</i> 2017 Barcelona
Automatic Design and Optimization of Deep Learning Models. <i>Rolando Garcia</i> , Randal Olson, Jason Moore. <i>The Leadership Alliance National Symposium</i> , Stamford, CT. July 2016.	<i>LANS</i> 2016
VAST 2015 Grand Challenge – Team VADER/VIS. Michael Steptoe, Robert Krüger, Yifan Zhang, Xing Liang, <i>Rolando Garcia</i> , Sagarika Kadambi, Wei Luo, Thomas Ertl, Ross Maciejewski. <i>IEEE Visual Analytics Science and Technology Conference</i> , Chicago, IL. October 2015.	<i>VAST</i> 2015 Chicago

TEACHING

Introduction to Database Systems (COMPSCI 186/286) http://cs186berkeley.net Graduate Student Instructor, 450 students. Platform: bCourses, Blackboard, Gradescope, Piazza	Spring 2018 <i>University of California, Berkeley</i>
Introduction to Programming Languages (CSE 240) Undergraduate Student Instructor, 267 students	Fall 2015 <i>Arizona State University, Tempe</i>
Introduction to Ethics (PHI 105) Undergraduate Student Instructor, 50 students	Fall 2013 <i>Northern Arizona University, Flagstaff</i>

SERVICE

Diversifying Access to Research in Engineering (DARE) Organizer dare.berkeley.edu	June 2020 – December 2022 <i>University of California, Berkeley</i>
EECS Graduate Admissions Graduate Student Representative eecs.berkeley.edu	Fall 2021 <i>University of California, Berkeley</i>
Data Systems and Foundations Seminar Coordinator dsf.berkeley.edu	Fall 2020 <i>University of California, Berkeley</i>
UC Berkeley / Stanford Graduate Pathways to STEM Peer Advising Co-Chair and Peer Advisor GPS Hyperlink	June 2017 – December 2019 <i>UC Berkeley & Stanford</i>